Timing and Tides: Exploring Trends in Women's Clubs

Jenna Graff, Benjamin Lim, Aria Whalen, Noah Braun, & Jaime Davila
Department of Statistics & Data Science, Department of Economics, St. Olaf College

INTRODUCTION

The goal of this project is to examine the role American women played in U.S. politics before gaining formal voting rights in 1920.

Civic engagement is examined through the formation of Women's Clubs (WCs). WCs were utilized as social and educational gatherings for women; they became places to advocate for reform through lobbying groups, provide support through mutual aid, and address social issues, such as education and child labor.

We aim to examine how social indicators like race, work status, and education predict the number of WCs formed across the United States between the Civil War and ratification of the 19th amendment (1865-1925).

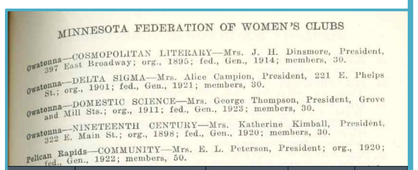
METHODS

Data for this project came from records of WCs printed by the General Federation of Women's Clubs (GFWC). Text data was extracted using Scantailor Advanced and Ocular Character Recognition (OCR) in R. Included variables were location of WCs (state, city), number of members in a WC, and the year of WC formation. Data was later combined with the Simple Maps U.S. Cities Data set and National Historical Geographic Information System (NHGIS) data sourced from the IPUMS database for interpretation.

To create an analysis of significant time periods in WC creation, historical events within our time period were compared against the number of WCs created per year, and relevant baselines. Historical events were the creation of the GFWC, the duration of WWI, and the ratification of the 19th amendment.

To analyze social factors that led to WC creation, we created a Hurdle model to predict if a county had one or more WCs, and if so, how many clubs.

Finally, number of WCs formed per year by state was visualized in R to reveal spatial trends.



MINNESOTA FEDERATION OF WOMEN'S CLUBS

Owatonna—COSMOPOLITAN LITERARY—Mrs. J. H. Dinsmore, President, 207 East Broadway; org., 1895; fed., Gen., 1914; members, 30.

Owatonna—DELTA SIGMA—Mrs. Alice Campion, President, 221 E. Phelps St.; org., 1901; fed., Gen., 1921; members, 30.

Owatonna—DOMESTIC SCIENCE—Mrs. George Thompson, President, Grove and Mill Sts.; org., 1911; fed., Gen., 1923; members, 30.

Owatonna—NINETEENTH CENTURY—Mrs. Katherine Kimball, President, 322 E. Main St.; org., 1898; fed., Gen., 1920; members, 30.

Pelican Rapids—COMMUNITY—Mrs. E. L. Peterson, President; org., 1920; fed., Gen., 1922; members, 50.

city_fixed	name	year_created	members	state
Owatonna	COSMOPOLLITAN LITERARY	1895	30	Minnesota
Owatonna	DELTA SIGMA	1901	30	Minnesota
Owatonna	DOMESTIC SCIENCE	1911	30	Minnesota
Owatonna	NINETEENTH CENTURY	1898	30	Minnesota
Pelican Rapids	COMMUNITY	1920	50	Minnesota
Pine Island	PROGRESS	1912	82	Minnesota

RESULTS

Figure 1: Number of WCs Created between 1865-1925

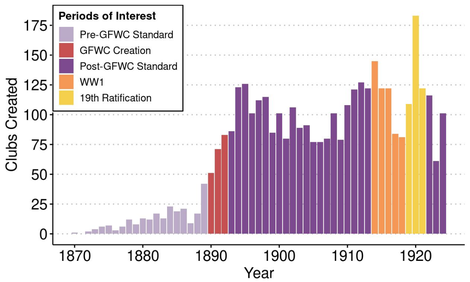


Table 1: ANOVA - Important Dates of WCs creation

Term Comparison	Difference	P-value adj.
Pre GFWC Standard vs. Post GFWC Standard	-87.01	0.000
GFCW Creation Period vs. Post GFCW Standard	-31.04	0.057
19th Amendment Period vs. Post GFWC Standard	38.63	0.010
World War 1 Period vs. Post GFCW Standard	11.43	0.710

Table 2: Logit Hurdle Model* - Odds of Having WCs

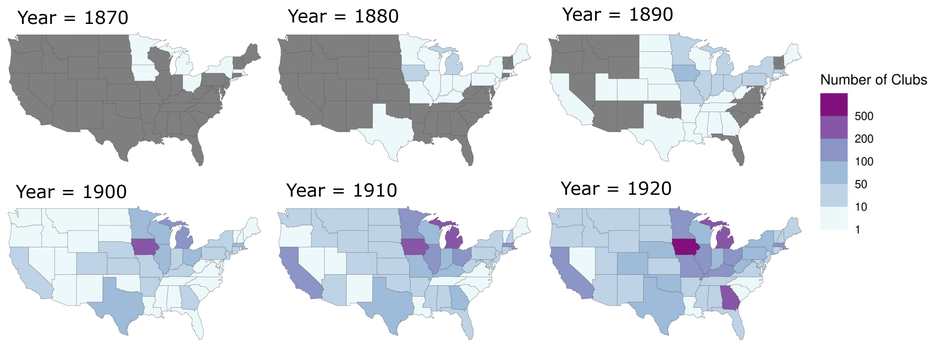
Term	Multiplicative Effect	P-value
Intercept	1.262	0.820
Urban Proportion	1.155	<0.001
White Proportion	0.950	0.135
Female Proportion	0.389	<0.001
Student Proportion	0.888	0.269
Adult Illiteracy Proportion	0.354	<0.001
Farmer Proportion	0.919	0.524
Women Farmer Proportion	0.412	<0.001
Log Total Population	x2 = 1.492	<0.001

Table 3: Count Hurdle Model* - Number of WCs

Term	Multiplicative Effect	P-value
Intercept	0.007	<0.001
Urban Proportion	0.904	<0.001
White Proportion	0.723	<0.001
Female Proportion	0.902	0.364
Student Proportion	1.636	<0.001
Adult Illiteracy Proportion	0.114	<0.001
Farmer Proportion	1.283	0.004
Women Farmer Proportion	0.421	<0.001

*Proportion estimates correspond with a 10% change

Figure 2: Map of Number of WCs per State between in 1870 and 1920**



* Mississippi, Florida, Montana, Colorado & Oklahoma underrepresented due to over 50% of WCs not reporting a creation date

INTERPRETATIONS

In Figure 1 & Table 1, we examine differences in number of WCs created over time. Notably, before and after the establishment of the GFWC, there is a baseline average difference of 87 clubs created every year. When the 19th amendment was ratified, there was an average increase in WCs created of about 39 clubs over 3 years when compared to the post GFWC standard.

From our hurdle model, interpretations of each exponentiated value affect either the likelihood of a county having a WC, or the expected number of clubs in a county (Tables 2 & 3).

- For every 10% increase in urban area in a county, the odds that the county has at least one WC are 1.155 times higher (Table 2).
- For counties with WCs, the expected number of clubs per capita to increase by 1.64 times for every 10% increase in the proportion of children who are students (Table 3).
- For counties with WCs, the expected number of clubs per capita to decrease is by 88.6% for every 10% increase in the proportion of adults who are illiterate (Table 3).

CONCLUSIONS

The first large increase in the number of WCs coincides with the creation of the GFWC. We also see large numbers of WCs in the Midwest, particularly Iowa, and tides of WCs spreading out over time. Counties with a lower proportion of adult illiteracy and a higher proportion of students had more WCs, while counties that were more urban and had a higher proportion of women were less likely to have any WCs. Overall, this analysis gives us insight into what kinds of social factors and historic events influenced civic engagement, and helps us understand patterns of women's political activity before gaining suffrage.

ACKNOWLEDGEMENTS

We would like to thank the Center for Interdisciplinary Research for giving us this opportunity and a special thank you to Paul Roback for assisting in model consultation.